

Songyao Xue

Luoyang, Henan, China |

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EDUCATION

Luoyang Normal University, B.Eng. in Software Engineering

Sept. 2023 – Present | Expected July 2027 | Fall 2027 Master's Applicant

- Academic Ranking: Top 10%.
- Relevant Coursework: Programming in C/Java/Python/JavaScript, Data Structures, Algorithm Design and Analysis, Database Systems (SQL), Probability and Statistics, Linear Algebra.
- English: CET-4 477; CET-6 first attempt 414; Mandarin Proficiency: Level 2-A.

RESEARCH INTERESTS

- Quantum information and quantum communication
- Quantum cryptography and quantum signature protocols
- Quantum games and multi-agent decision frameworks
- AI-assisted scientific computing, mathematical modeling, and Python/Qiskit simulations

RESEARCH EXPERIENCE

Quantum Information, Quantum Cryptography, and Quantum Game Theory

Undergraduate Research Assistant | Luoyang Normal University | 2023–Present

- Conducted research in quantum cryptography, quantum communication, quantum game theory, and probabilistic quantum decoding.
- Conducted literature review, mathematical modeling, theoretical derivation, and simulation-based validation.
- Used Python, Qiskit, PennyLane, and numerical experiments to analyze protocol behavior, noise robustness, and resource trade-offs.
- Prepared English manuscripts using LaTeX, including mathematical derivations, appendix proofs, figures, tables, and reference management.

RESEARCH TRAINING

Research Training in Quantum Information and Quantum Communication

Research Trainee / Supervisor-led Reading Group | Luoyang Normal University | 2023–Present

- Participated in weekly supervisor-led seminars and research discussions on quantum information, quantum communication, and quantum cryptography.
- Studied the theoretical foundations of quantum states, quantum circuits, quantum algorithms, quantum communication protocols, and quantum key distribution.
- Critically reviewed recent research papers and discussed protocol design, security analysis, and mathematical modeling techniques.
- Reinforced theoretical knowledge through Qiskit- and PennyLane-based programming exercises and numerical simulations.

MANUSCRIPTS AND PAPERS

On Grover-Based Quantum Signature Schemes

Second Author | Submitted to Entropy (JCR Q2) | Revised and resubmitted | 2025–2026

- Contributed to reusable criteria abstraction for quantum key distribution and quantum signature protocols.
- Supported mathematical framework design, formula organization, Qiskit-based verification, manuscript writing, LaTeX formatting, and reference management.

A Learnable Unified Framework for Weak-Link Multi-Group Quantum Games

First Author | Submitted to Quantum Machine Intelligence (JCR Q2) | Under Editorial Review | 2025–2026

- Proposed a weak-link multi-group quantum game framework using graph structures to model cooperation, competition, and weak coupling.
- Designed game tensors and symbolic rules, deriving first-order gain conditions, AND/OR task advantages, and spectral-radius constraints.
- Implemented Python-based automated search, batch experiments, noise analysis, training curves, network visualizations, and result tables.

Connectivity and Weak-Link Correlations in Multi-Group Entangled Quantum Networks

First Author | Manuscript in Preparation | planned submission in 2026 | 2026

- Developed a quantum communication framework based on multi-group entanglement, weak-link correlations, and no-signaling constraints.

- Currently finalizing notation consistency, formatting, and manuscript structure for submission.

Error-Corrected Quantum-Huffman-Inspired Probabilistic Source Decoding with Group Ancilla and Posterior Correction

First Author | Manuscript in Preparation | planned submission in 2026 | 2025–2026

- Constructed a probabilistic quantum Huffman-inspired decoding framework with grouped ancilla, QEC, and posterior correction.
- Used Python/Qiskit for noise scanning, group-gain analysis, QEC redundancy experiments, and resource phase-diagram visualization; introduced SDP-certified criteria for resource trade-off verification.

SELECTED PROJECTS

Youth Marxist Training Program Information Management System

School-Used Project | Independent Developer | May 2026

- Independently designed and developed a full-stack information management system for school-level training program administration.
- Built backend services and administrator/trainee/graduate-facing web pages with mobile-browser adaptation.
- Configured Ubuntu server deployment and packaged desktop clients for macOS and Windows using Electron.

Reflectance-Based Epitaxial Layer Thickness Estimation Using Multi-Beam Detection and Transfer Matrix Method

Mathematical Modeling Project | Core Developer / Team Leader | Sept. 2025

- Built mathematical, optical, and physical models for epitaxial layer thickness estimation.
- Applied FFT and the Transfer Matrix Method to process reflectance signals; implemented the pipeline in Python and used Bootstrap methods to estimate confidence intervals.

Archimedean Spiral-Based Optimization Design Model

Mathematical Modeling Project | Core Developer / Team Lead | Sept. 2024

- Modeled trajectories based on Archimedean spiral and parametric equations.
- Applied particle swarm optimization for global adjustment and numerical calculation; contributed to modeling, code implementation, and manuscript writing.

Financial Website Data Crawling and Cleaning System

Independent Python Project | Apr. 2024 – Jun. 2024

- Cleaned thousands of stock-related records using Pandas, handling missing values, outliers, and duplicates.
- Parsed HTML/XML using BeautifulSoup and used Selenium to simulate browser behavior for dynamic data retrieval.

TECHNICAL SKILLS

- Programming: Python, C, Java, JavaScript, SQL.
- Scientific Computing & Data Analysis: NumPy, Pandas, Matplotlib, Bootstrap methods, numerical simulation.
- Quantum Computing Tools: Qiskit, PennyLane.
- Software Development: Backend development, mobile adaptation, Electron, Ubuntu server deployment.
- Research Writing: LaTeX, Overleaf, TeXstudio, English manuscript writing, reference management.
- AI-assisted development workflows, code review and research productivity tools.

HONORS AND AWARDS

- Provincial Third Prize, Chinese Mathematics Competition, Henan Province, Nov. 2024
- Provincial Third Prize, Contemporary Undergraduate Mathematical Contest in Modeling, Henan Province, Sept. 2025
- Provincial Third Prize, Chinese Mathematics Competition, Henan Province, Nov. 2025
- Outstanding Student Cadre, Luoyang Normal University, Nov. 2025
- Merit Student Award, Luoyang Normal University, Nov. 2025
- Honorable Mention, Shuwei Cup Mathematical Modeling Challenge, Jan. 2026

LEADERSHIP AND SERVICE

Chief Administrator, Mathematical Modeling Laboratory *Sept. 2023 – Present*

- Organized regular mathematical modeling discussions, course teaching support, laboratory meetings, training activities, and project-based learning.

Committee Member, College Student Union *Sept. 2023 – Sept. 2024*

- Coordinated meeting tasks and supported the organization of more than five college-level student activities.

Management Committee Leader, Youth Service Group, University Youth League Committee *Mar. 2025 – Present*

- Managed student data for Youth Marxist Training Program members and supported duty scheduling and university-level activity coordination.

Volunteer, “San Xia Xiang” Summer Social Practice Program *Sept. 2025*

- Participated in community-oriented volunteer service and social practice activities; accumulated 106 volunteer service hours.